23K-0064
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Agentic AI Assignment (Beta)

Date _____

Question – 3

Given $z_t = [5, 3, 2, 0]$

$z_s = [3, 2.5, 1, 0.5]$

$y = [1, 0, 0, 0]$

$T = 2, \quad \alpha = 0.5, \quad \eta_1 = 0.5$

$L = \alpha T^2 KL(p_T \| q_T) + (1-\alpha) CE(y, q_1)$

(a) Teacher and student Probabilities at $T = 1$

→ softmax: $p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$

→ Teacher: $e^{z_t} = [148.413, 20.086, 7.389, 1]$

$\sum e^{z_t} = 176.888$

$p_1 = \frac{[148.413, 20.086, 7.389, 1]}{176.888}$

$\boxed{p_1 = [0.8390, 0.1135, 0.0418, 0.0057]}$

→ Student: $e^{z_s} = [20.086, 12.182, 2.718, 1.649]$

$\sum e^{z_s} = 36.635$

$q_1 = \frac{[20.086, 12.182, 2.718, 1.649]}{36.635}$

$\boxed{q_1 = [0.5483, 0.3325, 0.0742, 0.0450]}$

(b) Softened Probabilities at $T = 2$

Temperature Softmax: $p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$

→ Teacher: $z_t/2 = [2.5, 1.5, 1, 0]$

$e^{z_t/2} = [12.182, 4.482, 2.718, 1]$

$\sum e^{z_t/2} = 20.382$

$\boxed{p_T = [0.5977, 0.2199, 0.1334, 0.0491]}$

→ student: $z_s/2 = [1.5, 1.25, 0.5, 0.25]$

$e^{z_s/2} = [4.482, 3.490, 1.649, 1.284]$

$\sum e^{z_s/2} = 10.905$

$q_T = \frac{[4.482, 3.490, 1.649, 1.284]}{10.905}$

$\boxed{q_T = [0.4110, 0.3201, 0.1512, 0.1177]}$

(c) KL Divergence:

$KL(p_T \| q_T) = \sum_i p_{T,i} \ln\left(\frac{p_{T,i}}{q_{T,i}}\right)$

→ $KL = 0.5977 \ln(0.5977/0.4110) + 0.2199 \ln(0.2199/0.3201) + 0.1334 \ln(0.1334/0.1512) + 0.0491 \ln(0.0491/0.1177)$

$KL = 0.2239 - 0.0826 - 0.0167 - 0.0430$

$\boxed{KL = 0.0816}$

(d) Hard-label cross-entropy loss

$CE = (y, q_1) = -\sum y_i \ln(q_{1,i})$

$CE = -[1 \ln(0.5483) + 0 + 0 + 0]$

$\boxed{CE = 0.6010}$

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(e) Total Distillation Loss

$$L = \alpha T^2 KL + (1-\alpha) CE$$
$$= 0.5(2)^2 (0.0816) + 0.5 (0.6010)$$
$$\boxed{L = 0.4637}$$

(f) Gradient w.r.t Student Logits

→ KL gradient : $\frac{\partial KL}{\partial z_s} = \frac{q_T - p_T}{T}$

→ After temperature scaling : $\alpha T (q_T - p_T)$

→ cross-entropy gradient : $(1-\alpha)(q_1 - y)$

→ Combined gradient.

$$\nabla L = \alpha T (q_T - p_T) + (1-\alpha)(q_1 - y)$$

→ KL component = $0.5(2)(q_T - p_T)$

$$= [0.1867, 0.1002, 0.0178, 0.0687]$$

→ Cross entropy component = $0.5(q_1 - y)$

$$= [-0.2259, 0.1663, 0.0371, 0.0225]$$

Total Gradient :

Word	KL Gradient	CE Gradient
hospital	−0.1867	−0.2259
clinic	0.1002	0.1663
ward	0.0178	0.0371
school	0. 0.687	0.0225

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$$\boxed{\nabla L = [-0.4126, 0.2665, 0.0549, 0.0912]}$$

(g) Updated Student Logits and New Loss

Gradient descent update :

$$z_s^{new} = z_s - \eta \nabla L$$
$$= [3, 2.5, 1, 0.5] - 0.5 [-0.4126, 0.2665, 0.0549, 0.091$$
$$\boxed{z_s^{new} = [3.2063, 2.3668, 0.9725, 0.4544]}$$

New probabilities at $T=1$

$$q_1^{new} = [0.6239, 0.2695, 0.0868, 0.0398]$$

New probabilities at $T=2$

$$q_T^{new} = [0.4470, 0.2938, 0.1463, 0.1129]$$

New KL Divergence : $KL^{new} = 0.0567$

New cross-entropy : $CE^{new} = -\ln(0.6239) = 0.4718$

New Total loss : $L^{new} = 0.5(2)^2 (0.0567) + 0.5(0.471$

$$\boxed{L^{new} = 0.3493}$$

Comparison $L_{before} = 0.4637$

$L_{after} = 0.3493$

→ loss decreased by approximately 24.7 %.

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Question-4

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(1) Zero-shot accuracy and 95% confidence interval

→ Correct predictions: TP = 100 + 80 + 50 = 230

Accuracy = $\frac{230}{300}$

Accuracy = 0.7667

Standard error ⇒ $SE = \sqrt{\frac{P(1-P)}{n}}$

$= \sqrt{(0.7667(1-0.7667))/300}$

SE = 0.0244

95% confidence interval: $CI = P \pm 1.96(SE)$

$CI = 0.7667 \pm 1.96(0.0244)$

CI = [0.7188, 0.8145]

(2) Precision, Recall, F1 for each class (Zero-shot)

Precision = $\frac{TP}{\text{Column Total}}$, Recall = $\frac{TP}{\text{Row Total}}$, $F1 = \frac{2P.R}{P+R}$

→ Positive: $P = \frac{100}{128} = 0.7813$

$R = \frac{100}{120} = 0.8333$

$F1 = \frac{2(0.7813)(0.8333)}{0.7813 + 0.8333}$; **F1 = 0.8065**

→ Neutral: $P = \frac{50}{77} = 0.6494$

$R = \frac{50}{80} = 0.6250$

$F1 = \frac{2(0.6494)(0.6250)}{0.6494 + 0.6250}$ ⇒ **F1 = 0.6369**

→ [illegible] = [illegible] + 0.8205 + 0.6369 / 3

[illegible] = [illegible]

→ Negative: $P = 80/95 = 0.8421$

$R = 80/100 = 0.8000$

$F1 = \frac{2(0.8421)(0.8000)}{0.8421 + 0.8000}$; **F1 = 0.8205**

(3) Macro, Weighted and Micro F1 (Zero-shot)

→ Macro F1 = (0.8065 + 0.8205 + 0.6369) / 3

Macro F1 = 0.7546

→ Weighted F1 = $\frac{120(0.8065) + 100(0.8205) + 80(0.6369)}{300}$

Weighted F1 = 0.7659

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→ Micro F1 : TP = 230, FP = 70, FN = 70

Micro P = 230/300 = 0.7667

Micro R = 230/300 = 0.7667

Micro F1 = 0.7667

(4) Stage 1 Interpretation and Recommendations

- Neutral has the lowest recall (62.5%) & precision (64.9%)
- 20 Neutral reviews were incorrectly predicted as positive. 15 positive reviews were predicted as Neutral, while 12 negative reviews were also predicted as neutral.
- positive & negative are less confused with each other (58 cases)
- The main issue is the classification of neutral reviews
- Since the prompt only contains label names, the model does not have a clear definition of Neutral. It tends to classify mixed reviews based on the opinion expressed first.

Recommendations:

1) Provide a clear definition of Neutral (factual, mixed, or no clear opinion)
2) Include few-shot examples for all three classes, especially borderline mixed reviews
3) Ask the model to identify both positive and negative points before assigning a label
4) Restrict the output to the three available labels

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(5) Stage 2 Evaluation & Comparison with Stage 1

→ Accuracy = (106 + 84 + 64) / 300

Accuracy = 0.8467

→ Standard error : $SE = \sqrt{(0.8467(1-0.8467))/300}$

SE = 0.0208

→ 95% confidence interval:

$CI = 0.847 \pm 1.96\ (0.0208)$

CI = [0.8059, 0.8874]

→ Positive : $P = 106/121 = 0.8760$, $R = 106/120 = 0.8833$

$F1 = \frac{2(0.8760)(0.8833)}{0.876 + 0.8833}$ => **F1 = 0.8797**

→ Negative : $P = 84/95 = 0.8842$, $R = 84/100 = 0.8400$

$F1 = \frac{2(0.8842)(0.8400)}{0.8842 + 0.8400}$, **F1 = 0.8615**

→ Neutral : $P = 64/84 = 0.7619$, $R = 64/86 = 0.800$

F1 = 0.7805

→ Macro F1 : (0.8797 + 0.8615 + 0.7805)/3 =

Macro F1 = 0.8406

→ Weighted $F1 = \frac{120(0.8797) + 100(0.8615) + 80(0.7805)}{300}$

Weighted F1 = 0.8472

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→ Micro F1 : TP = 254 , FP = 46 , FN = 46

Micro P : 254/300 = 0.8469

Micro R : 254/300 = 0.8469

Micro F1 = 0.8464

Comparison

Metric	Stage 1	Stage 2	Change
Accuracy	0.7667	0.8467	+0.0800
Macro F1	0.7546	0.8406	+0.0859
Weighted F1	0.7659	0.8472	+0.0812
Micro F1	0.7667	0.8467	+0.0800
F1 Neutral	0.6769	0.7805	+0.1435

All metrics improved in Stage 2. The biggest improvement was in Neutral, where recall increased 62.5% to 80%. Neutral reviews incorrectly classified as positive decreased from 20 to 9. The revised definition and few-shot examples improved the classification of mixed reviews. The two confidence intervals overlap slightly (0.8145 and 0.8059), supporting the observed improvement.

(6) Prompt Stability

→ Mean $\bar{x} = (0.847 + 0.840 + 0.853 + 0.837 + 0.850) / 5$

$\bar{x} = 0.8454$

→ Deviations from mean : [0.0016, −0.0054, 0.0076, −0.0084, 0.0046]

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→ Squared Deviations : [0.00000256, 0.00002916, 0.00005776, 0.00007056, 0.00002116]

$\sum (x_i - \bar{x})^2 = 0.0001812$

→ Sample Standard Deviation : $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

$= \sqrt{0.0001812 / 4}$

$s = 0.0067$

→ Final Result : 0.8454 ± 0.0067

The prompt is stable. Accuracy ranges from 83.7% to 85.3%, a difference of only 1.6 percentage points. The ordering of few-shot examples has little effect on the results.

(7) BLEU-4, ROUGE-1, ROUGE-2, ROUGE-L & METEOR

→ Reference = 10 tokens , Candidate = 9 tokens

→ BLEU-4 : Modified n-gram precisions :

$p_1 = 8/9 = 0.8889$, $p_2 = 5/8 = 0.6250$

$p_3 = 3/7 = 0.4286$, $p_4 = 2/6 = 0.3333$

Brevity penalty : $BP = e^{(1 - r/c)}$

$= e^{(1 - 10/9)}$

$BP = 0.8948$

BLEU-4 : $BLEU = BP \times \exp\left(1/4 \cdot \sum_{n=1}^{4} \ln p_n\right)$

$= 0.8948 \times \exp\left(\frac{-0.1178 - 0.4700 - 0.8473 - 1.0986}{4}\right)$

BLEU-4 = 0.4750

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★ ROUGE-1 : overlap = 8

$P = 8/9 = 0.8889$, $R = 8/10 = 0.800$

$F1 = \frac{2(0.8889)(0.800)}{0.8889 + 0.800}$

ROUGE-1 = 0.8421

★ ROUGE-2 : overlap = 5

$P = 5/8 = 0.6250$, $R = 5/9 = 0.5556$

$F1 = \frac{2(0.625)(0.5556)}{0.625 + 0.5556}$

ROUGE-2 = 0.5882

★ ROUGE-L LCS = 5

$P = 5/9 = 0.5556$, $R = 5/10 = 0.5$

$F_1 = [2(0.556)(0.5)] / [0.556 + 0.5)$

ROUGE-L = 0.5263

★ METEOR

$Fmean = \frac{PR}{0.9P + 0.1(R)}$

$= \frac{(0.8889)(0.5)}{0.9(0.8889) + 0.1(0.8)}$

fmean = 0.8081

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★ Fragmentation Penalty : chunks = 3

$Penalty = 0.5 \times (3/8)^3$

Penalty = 0.0264

★ METEOR : $METEOR = Fmean \times (1 - Penalty)$

$= 0.8081 (1 - 0.0264)$

METEOR = 0.7868

Interpretation : Both sentences have the same meaning, but their word order differs and "late" is replaced with "slow". ROUGE-1 & METEOR are higher due to word overlap, wh BLEU, ROUGE-2 & ROUGE-L are lower due to fewer matching n-grams. BERTScore is more suitable for capturing semantic similarity.

⑧ Perplexity

→ Given token probabilities:

$P = [0.60, 0.40, 0.50, 0.80, 0.90, 0.30, 0.25, 0.50, 0.20$

→ negative log probabilities:

$\sum -\ln(p_i) = 7.5929$

→ average negative log probability : $7.5929/9 = 0.843$

→ calculate perplexity : $PPL = e^{0.8437}$

PPL = 2.3249

⇒ Model is about as uncertain as choosing b/w 2.3 ea likely tokens at each step. Its confidence is lowest for " (0.20), "delivery" (0.25) and "but" (0.30). Perplexity confidence, not correctness.